

JKBOSE 2026

CLASS 11TH

STRAIGHT LINES

MATHS

LECTURE- 07

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TOPICS TO BE COVERED

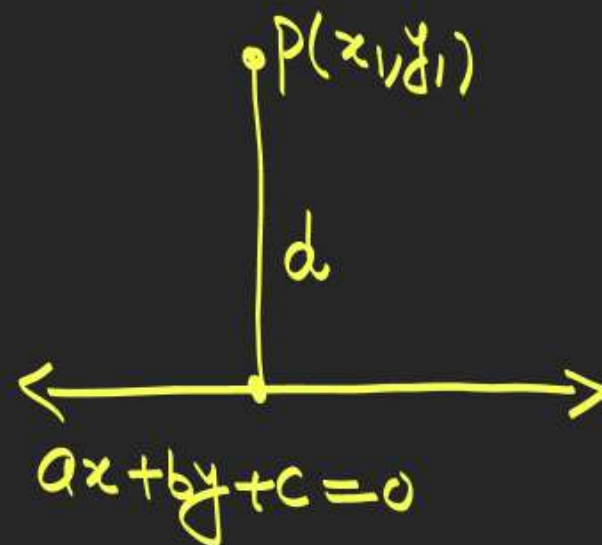
1

Exercise Discussion 9.3

2

Related Problems

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$



$$\begin{aligned} & \text{Line 1: } ax + by + c' = 0 \\ & \text{Line 2: } ax + by + c = 0 \\ & \text{Distance } d = \frac{|c' - c|}{\sqrt{a^2 + b^2}} \end{aligned}$$

EXERCISE – 9.3

QUESTION

Reduce the following equations into slope – intercept form and find their slopes and the y -intercepts.

(i) $x + 7y = 0$ ✓

(ii) $6x + 3y - 5 = 0$ ✓

(iii) $y = 0(0 \cdot x + 0)$ ✓

(Sol)(i) Given equation is:

$$x + 7y = 0$$

$$\Rightarrow 7y = -x$$

$$\Rightarrow y = \left(-\frac{1}{7}\right)x + 0$$

$$y = mx + c$$

So, Slope = $-\frac{1}{7}$

y -intercept = 0.

(ii) Given that $6x + 3y - 5 = 0$

$$\Rightarrow 3y = -6x + 5$$

$$\Rightarrow y = -\frac{6x}{3} + \frac{5}{3}$$

$$\Rightarrow y = -2x + \frac{5}{3}$$

So, Slope = -2

y -intercept = $\frac{5}{3}$

QUESTION

$$\frac{x}{a} + \frac{y}{b} = 1$$

Reduce the following equations into intercept form and find their intercepts on the axes.

(i) $3x + 2y - 12 = 0$

(ii) $4x - 3y = 6$ (HP/W)

(iii) $3y + 2 = 0$ (HP/W)

10 (Sol) Given equation is:

$$3x + 2y - 12 = 0$$

$$\Rightarrow 3x + 2y = 12$$

$$\Rightarrow \frac{3x}{12} + \frac{2y}{12} = 1$$

$$\frac{x}{4} + \frac{y}{6} = 1$$
 which is the required intercept form.

with x -intercept = 4 & y -intercept = 6.

Hence the result.

QUESTION

Find the distance of the point $(-1, 1)$ from the line $12(x + 6) = 5(y - 2)$.

(Sol) Given line is:

$$12(x + 6) = 5(y - 2)$$

$$\Rightarrow 12x + 72 = 5y - 10$$

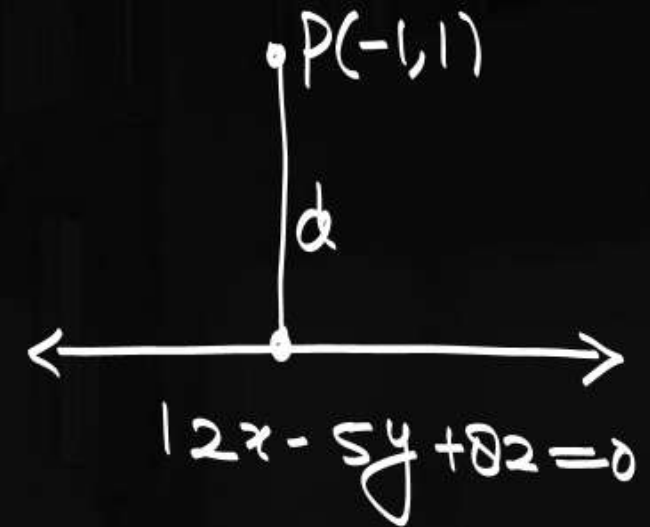
$$\Rightarrow 12x - 5y + 82 = 0 \quad \text{--- (1)}$$

Therefore the distance of $P(-1, 1)$ from (1) is:

$$d = \frac{|12(-1) - 5(1) + 82|}{\sqrt{12^2 + 5^2}} = \frac{|65|}{\sqrt{169}}$$

$$\Rightarrow d = \frac{65}{13} = \frac{5 \times \cancel{13}}{\cancel{13}}$$

$$d = 5 \text{ units.}$$



QUESTION

Find the points on the x -axis whose distances from the line $\frac{x}{3} + \frac{y}{4} = 1$ are 4 units.

(Sol) Given line is:

$$\frac{x}{3} + \frac{y}{4} = 1$$

$$\Rightarrow 4x + 3y = 12$$

$$\Rightarrow 4x + 3y - 12 = 0 \quad \text{--- (1)}$$

Let the required point on the x -axis be $P(a, 0)$ such that the distance of $P(a, 0)$ from (1) is 4 units.

$$\Rightarrow \frac{|4a + 3(0) - 12|}{\sqrt{4^2 + 3^2}} = 4$$

$$\Rightarrow \frac{|4a - 12|}{5} = 4$$

$$\Rightarrow |4a - 12| = 20$$

$$\Rightarrow 4a - 12 = 20, -20$$

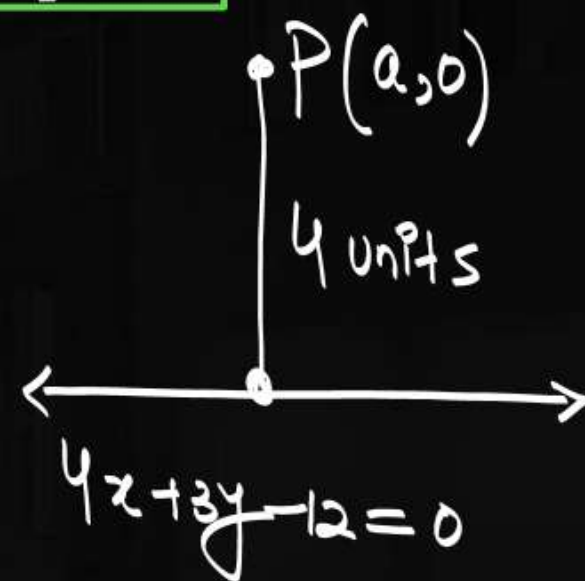
$$\Rightarrow 4a = 20 + 12, -20 + 12$$

$$\Rightarrow 4a = 32, -8$$

$$\Rightarrow a = 8, -2$$

Therefore the required points are $(8, 0)$ & $(-2, 0)$

Hence the result.



QUESTION

Find the distance between parallel lines.

(i) $15x + 8y - 34 = 0$ and $15x + 8y + 31 = 0$ (H/W)

(ii) $l(x + y) + p = 0$ and $l(x + y) - r = 0$

(Sol) (i) Given parallel lines are:

$$15x + 8y - 34 = 0 \quad \text{--- (1)}$$

$$15x + 8y + 31 = 0 \quad \text{--- (2)}$$

So the distance d b/w (1) & (2) is given by;

$$d = \frac{|-34 - 31|}{\sqrt{15^2 + 8^2}}$$

$$\Rightarrow d = \frac{|p+r|}{\sqrt{l^2 + l^2}} = \frac{|p+r|}{\sqrt{2}l} = \frac{|p+r|}{\sqrt{2}l}$$

$$\Rightarrow d = \frac{|p+r|}{\sqrt{2}l} \quad (p > 0, r > 0, l > 0)$$

#

QUESTION

Find equation of the line parallel to the line $3x - 4y + 2 = 0$ and passing through the point $(-2, 3)$.

$$y - y_1 = m(x - x_1)$$

$$\Rightarrow y - 3 = \frac{3}{4}(x + 2)$$

$$4y - 12 = 3x + 6$$

$$\Rightarrow 3x - 4y + 18 = 0. \checkmark \checkmark$$

$$3x - 4y + 2 = 0$$



$(-2, 3)$

$$m = -\frac{3}{4} = \frac{3}{4}$$

QUESTION

Find equation of the line perpendicular to the line $x - 7y + 5 = 0$ and having x intercept 3.

(Sol) Given line is:

$$x - 7y + 5 = 0 \quad \text{--- (1)}$$

Since the required line has x -intercept 3

So $(3, 0)$ lies on the required line.

Also, let m be the slope of the required line.

$$\text{So, } m \times \left(\frac{-1}{-7} \right) = -1$$

$$\Rightarrow \boxed{m = -7}$$

So, the required line is:

$$y - 0 = -7(x - 3)$$

$$\Rightarrow y = -7x + 21$$

$$\Rightarrow \boxed{7x + y - 21 = 0}$$

Hence the result.

QUESTION

(PYQ JKBOSE)

Find angles between the lines $\sqrt{3}x + y = 1$ and $x + \sqrt{3}y = 1$.

(Sol) Given lines are:

$$\sqrt{3}x + y = 1 \text{ --- ①}$$

$$x + \sqrt{3}y = 1 \text{ --- ②}$$

Here $m_1 = \text{slope of ①} = -\frac{1}{\sqrt{3}}$

$$\Rightarrow m_1 = -\frac{1}{\sqrt{3}}$$

Also, $m_2 = \text{slope of ②} = -\frac{1}{\sqrt{3}}$

$$\Rightarrow m_2 = -\frac{1}{\sqrt{3}}$$

Let θ be the acute angle b/w ① & ②,

$$\text{So, } \tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

$$\Rightarrow \tan \theta = \left| \frac{-\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}}}{1 + \left(-\frac{1}{\sqrt{3}}\right)\left(-\frac{1}{\sqrt{3}}\right)} \right|$$

$$\Rightarrow \tan \theta = \left| \frac{\cancel{-\frac{1}{\sqrt{3}}} + \cancel{\frac{1}{\sqrt{3}}}}{\frac{1}{2}} \right| = \left| \frac{\cancel{2}}{\cancel{2}\sqrt{3}} \right|$$

$$\Rightarrow \tan \theta = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta = \frac{\pi}{6} = 30^\circ$$



Also the other angle b/w ① & ②
 $= \pi - \frac{\pi}{6} = \frac{5\pi}{6}$ or 150° .

$\Rightarrow$ Required angles b/w ① & ②
are $\frac{\pi}{6}, \frac{5\pi}{6}$ or $30^\circ, 150^\circ$.

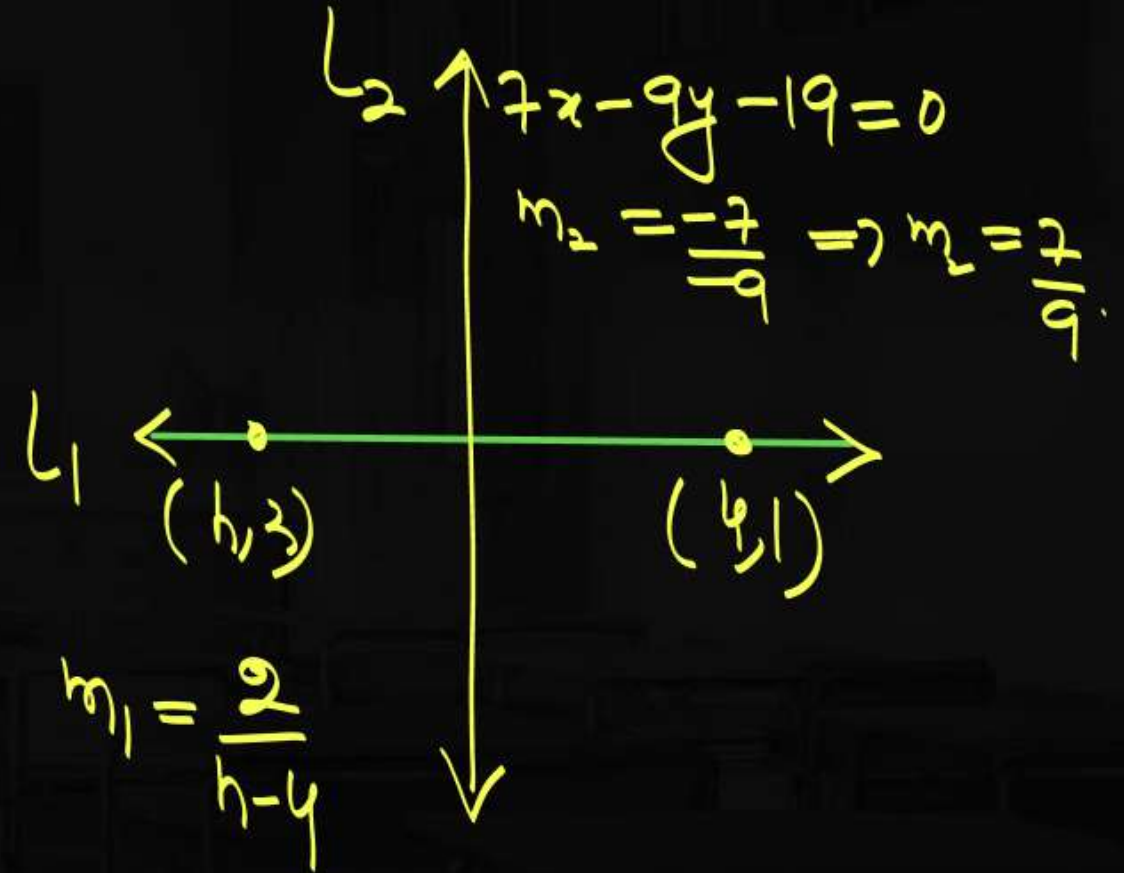
Hence the result.

QUESTION

The line through the points $(h, 3)$ and $(4, 1)$ intersects the line $7x - 9y - 19 = 0$ at right angle. Find the value of h .

$$\frac{7}{9} \times \frac{2}{h-4} = -1$$

H/W



QUESTION

Prove that the line through the point (x_1, y_1) and parallel to the line $Ax + By + C = 0$ is

$$A(x - x_1) + B(y - y_1) = 0$$

(Sol) let m be the slope of the required line.

$$\text{So, } m = -\frac{A}{B}$$

Therefore the required line is:

$$y - y_1 = -\frac{A}{B}(x - x_1)$$

$$\Rightarrow B(y - y_1) = -A(x - x_1)$$

$$A(x - x_1) + B(y - y_1) = 0$$

Hence proved.

QUESTION

Two lines passing through the point $(2, 3)$ intersects each other at an angle of 60° .

If slope of one line is 2, find equation of the other line.

(Sol) we know that

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

$$\Rightarrow \tan 60^\circ = \left| \frac{m - 2}{1 + 2m} \right|$$

$$\Rightarrow \sqrt{3} = \left| \frac{m - 2}{2m + 1} \right|$$

$$\Rightarrow \left| \frac{m - 2}{2m + 1} \right| = \sqrt{3}$$

$$\Rightarrow \frac{m - 2}{2m + 1} = \pm \sqrt{3}$$

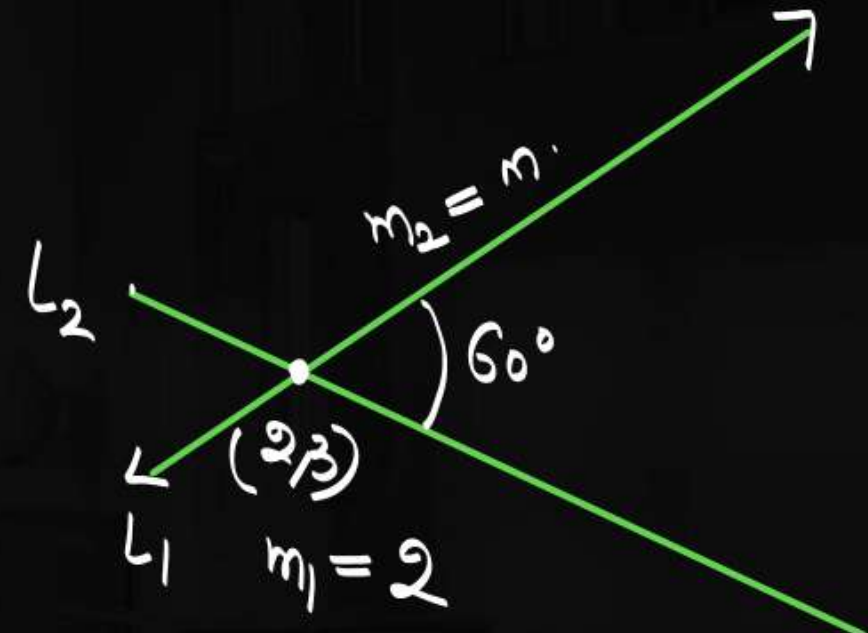
Case I: $\frac{m - 2}{2m + 1} = \sqrt{3}$

$$\Rightarrow m - 2 = 2\sqrt{3}m + \sqrt{3}$$

$$\Rightarrow m - 2\sqrt{3}m = \sqrt{3} + 2$$

$$\Rightarrow m(1 - 2\sqrt{3}) = \sqrt{3} + 2$$

$$\Rightarrow m = \frac{2 + \sqrt{3}}{1 - 2\sqrt{3}}$$



So, the required equation is:

$$y-3 = \left(\frac{2+\sqrt{3}}{1-2\sqrt{3}} \right) (x-2)$$

$$\Rightarrow (1-2\sqrt{3})y - 3 + 6\sqrt{3} = (2+\sqrt{3})x - 4 - 2\sqrt{3}$$

$$\Rightarrow (2+\sqrt{3})x + (2\sqrt{3}-1)y = 8\sqrt{3}+1$$

Case 2nd: $\frac{m-2}{2m+1} = -\sqrt{3}$

$$\Rightarrow m-2 = -2\sqrt{3}m - \sqrt{3}$$

$$\Rightarrow m(1+2\sqrt{3}) = 2-\sqrt{3}$$

$$\Rightarrow m = \frac{2-\sqrt{3}}{1+2\sqrt{3}}$$

So, the required equation is:

$$y-3 = \left(\frac{2-\sqrt{3}}{1+2\sqrt{3}} \right) (x-2)$$

$$\Rightarrow (1+2\sqrt{3})y - 3 - 6\sqrt{3} = (2-\sqrt{3})x - 4 + 2\sqrt{3}$$

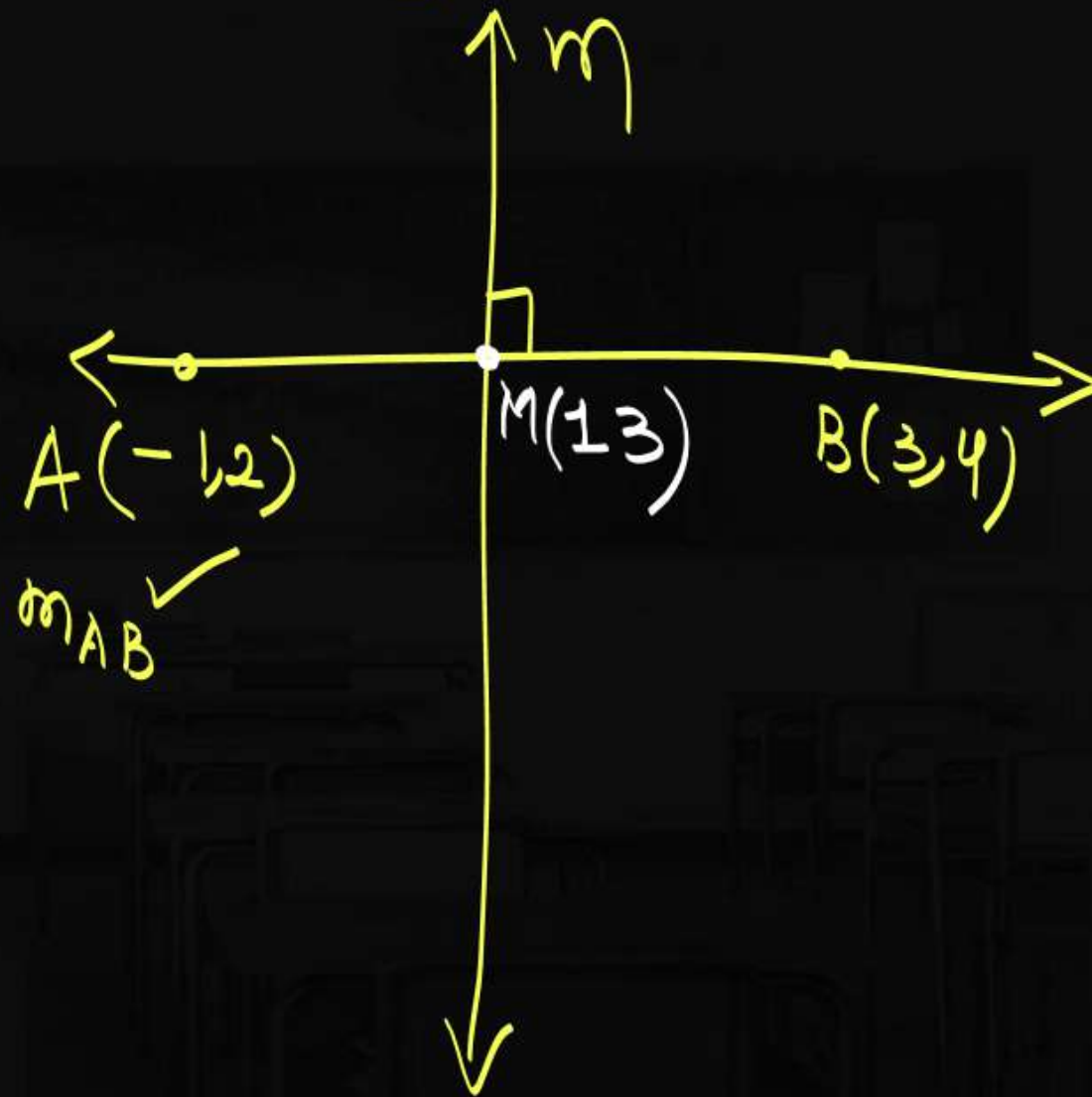
$$\Rightarrow (2-\sqrt{3})x - (2\sqrt{3}+1)y = 1-8\sqrt{3}$$

Hence the result

QUESTION

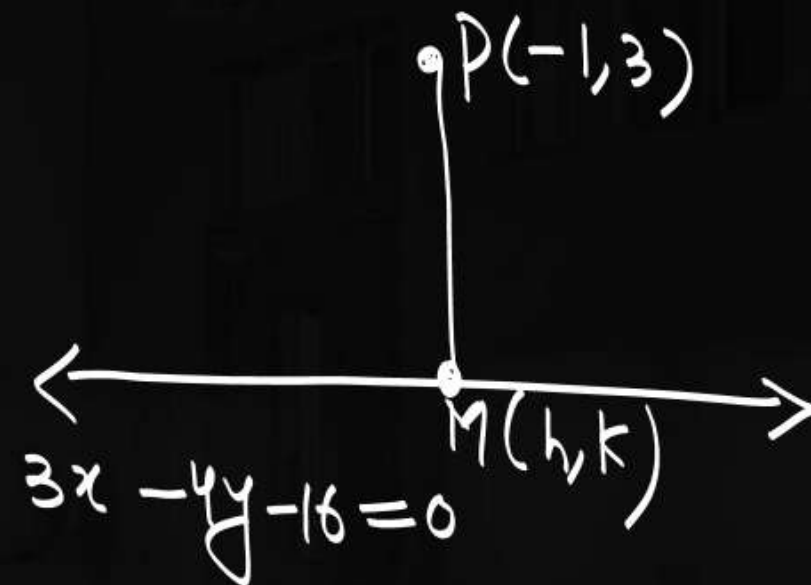
Find the equation of the right bisector of the line segment joining the points $(3, 4)$ and $(-1, 2)$. (H/W)

$$\begin{aligned} \text{If } x m_{AB} &= -1 \\ \Rightarrow m &= -\frac{1}{m_{AB}} \end{aligned}$$



QUESTION

Find the coordinates of the foot of perpendicular from the point $(-1, 3)$ to the line $3x - 4y - 16 = 0$. (H/W). (Already done).



QUESTION

$$y = \frac{1}{2}x + \frac{5}{2}$$

The perpendicular from the origin to the line $y = mx + c$ meets it at the point $(-1, 2)$.

Find the values of m and c .

(Sol) Since $OP \perp L$, therefore product of slopes $= -1$.

$$\Rightarrow m \times \left(\frac{2-0}{-1-0} \right) = -1$$

$$\Rightarrow m \times \frac{2}{-1} = -1$$

$$\Rightarrow m = \frac{1}{2}$$

Also, $(-1, 2)$ lies on L , so

$$2 = m(-1) + c$$

$$\Rightarrow 2 = -m + c$$

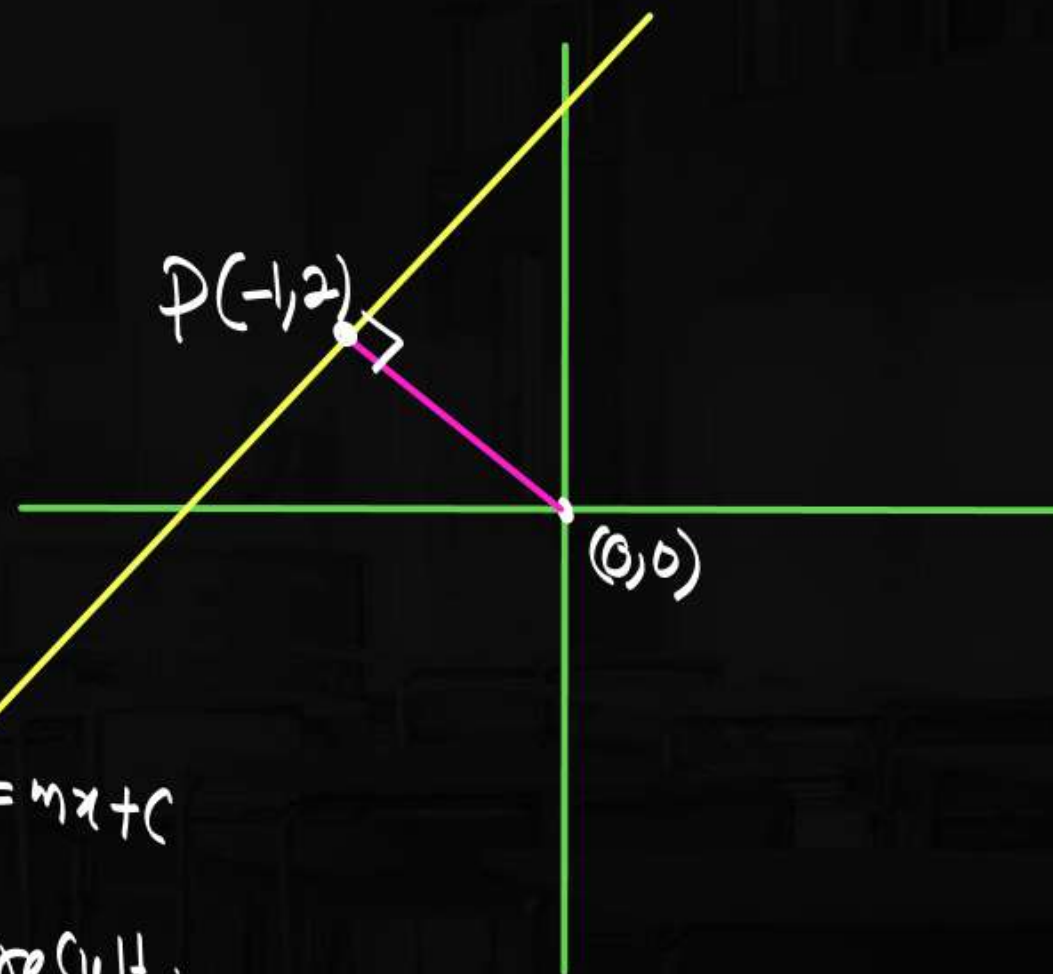
$$\Rightarrow c = 2 + m$$

$$\Rightarrow c = 2 + \frac{1}{2}$$

$$\Rightarrow c = \frac{5}{2}$$

$$L: y = mx + c$$

Hence the result.



QUESTION

$$p^2 + 4q^2 = k^2.$$

If p and q are the lengths of perpendiculars from the origin to the lines $x \cos \theta - y \sin \theta = k \cos 2\theta$ and $x \sec \theta + y \operatorname{cosec} \theta = k$, respectively, prove that $p^2 + 4q^2 = k^2$.

Proof: Given lines are:

$$x \cos \theta - y \sin \theta - k \cos 2\theta = 0 \quad (1)$$

$$x \sec \theta + y \operatorname{cosec} \theta - k = 0 \quad (2)$$

$$\text{So, } p = \frac{|0 \times \cos \theta - 0 \times \sin \theta - k \cos 2\theta|}{\sqrt{\cos^2 \theta + \sin^2 \theta}}$$

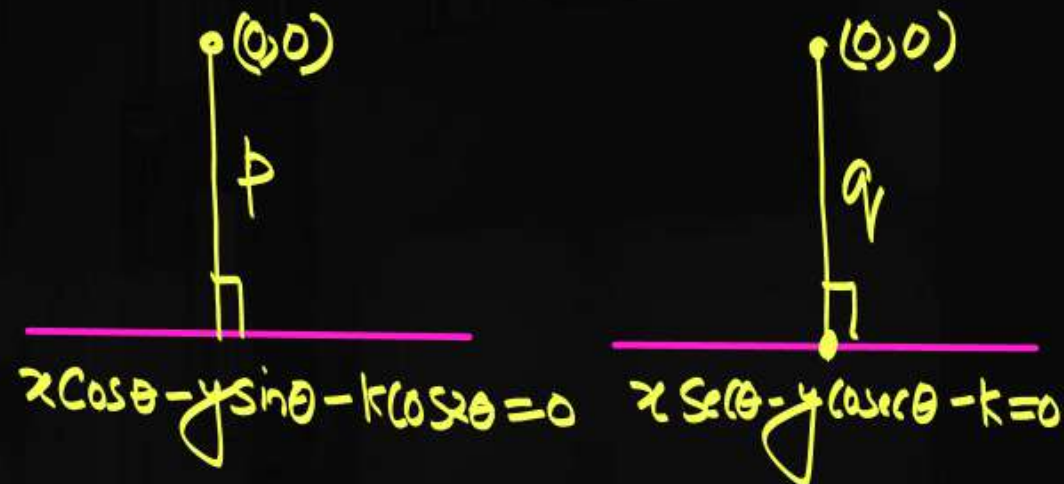
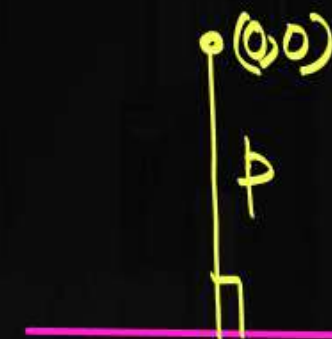
$$\Rightarrow p = \frac{|-k \cos 2\theta|}{\sqrt{1}}$$

$$\Rightarrow p = |-k \cos 2\theta|$$

$$\Rightarrow p^2 = k^2 \cos^2 2\theta \quad (1)$$

$$\text{Also } q = \frac{|0 \times \sec \theta - 0 \times \operatorname{cosec} \theta - k|}{\sqrt{\sec^2 \theta + \operatorname{cosec}^2 \theta}}$$

$$\Rightarrow q = \frac{|-k|}{\sqrt{\frac{1}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}}} = \frac{|-k|}{\sqrt{\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cos^2 \theta}}} = \frac{|-k|}{\sqrt{\frac{1}{\sin^2 \theta \cos^2 \theta}}}$$



$$\Rightarrow q = |-k| \sqrt{(\sin\theta \cos\theta)^2}$$

$$\Rightarrow q = |-k| |\sin\theta \cos\theta|$$

$$\Rightarrow q^2 = k^2 (\sin^2\theta \cos^2\theta)$$

$$\Rightarrow 4q^2 = k^2 \times 4(\sin\theta \cos\theta)^2$$

$$\Rightarrow 4q^2 = k^2 (2\sin\theta \cos\theta)^2$$

$$\Rightarrow 4q^2 = k^2 \sin^2 2\theta \quad \text{--- (2)}$$

Adding (1) & (2), we get;

$$p^2 + 4q^2 = k^2 \cos^2 2\theta + k^2 \sin^2 2\theta$$

$$\Rightarrow p^2 + 4q^2 = k^2 (\cos^2 2\theta + \sin^2 2\theta)$$

$$\Rightarrow p^2 + 4q^2 = k^2$$

Hence proved.

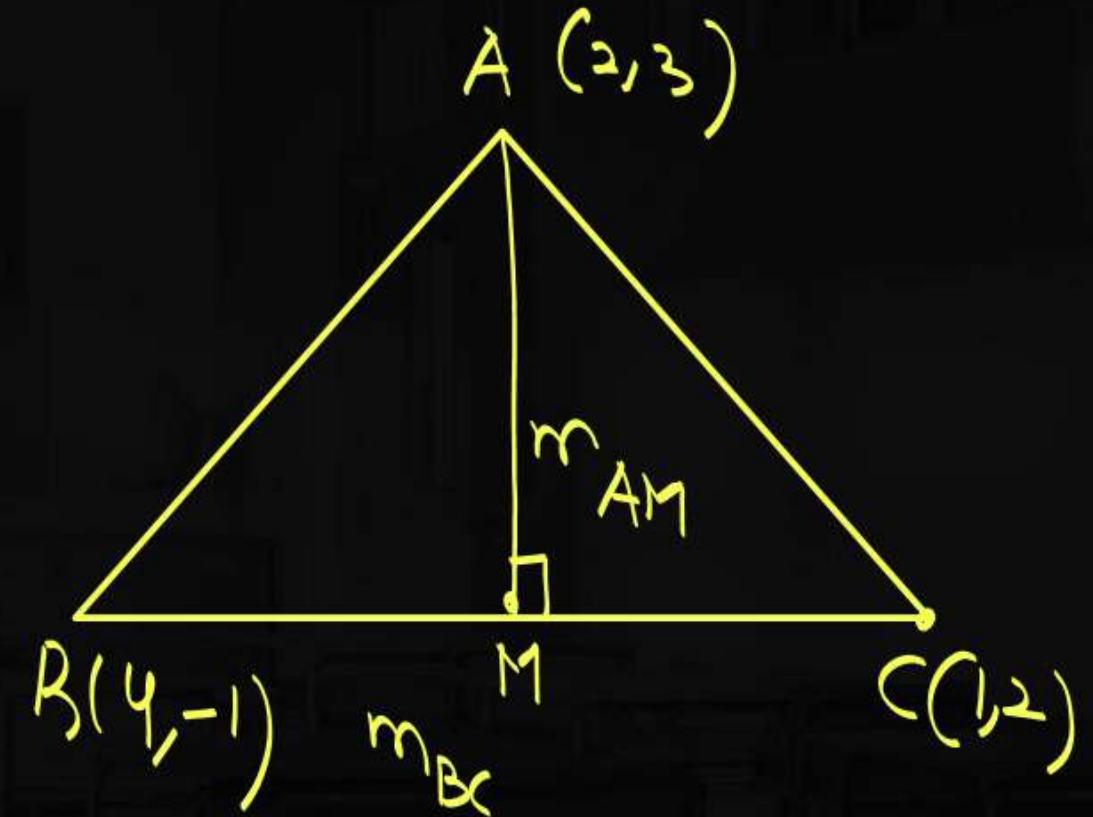
QUESTION

In the triangle ABC with vertices A (2, 3), B(4, - 1) and C(1, 2) find the equation and length of altitude from the vertex A. (Homework) Problem.

$$\checkmark$$

$$m_{AM} \times m_{BC} = -1.$$

$$y - y_1 = m_{AM}(x - x_1)$$



QUESTION

(Imp / JKBOSE)



If p is the length of perpendicular from the origin to the line whose intercepts on the axes are a and b , then show that $\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$.

Proof: Suppose that the equation of the given line is $\frac{x}{a} + \frac{y}{b} = 1$

$$\Rightarrow bx + ay = ab$$

$$\Rightarrow bx + ay - ab = 0 \quad \text{--- (1)}$$

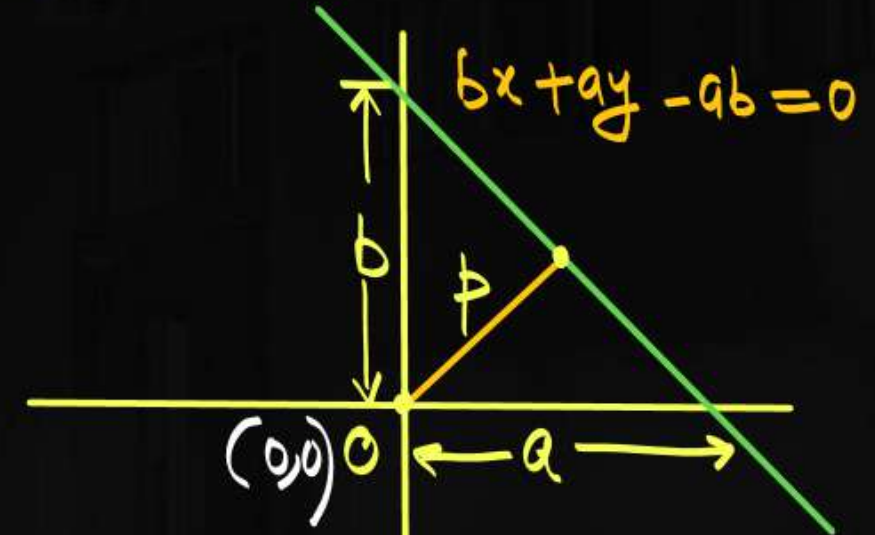
Since p is the length of the perpendicular from the origin to the line (1).

$$\text{So, } p = \frac{|bx_0 + ay_0 - ab|}{\sqrt{b^2 + a^2}}$$

$$\Rightarrow p = \frac{|ab|}{\sqrt{b^2 + a^2}}$$

$$\Rightarrow p^2 = \frac{a^2 b^2}{b^2 + a^2}$$

$$\Rightarrow \frac{1}{p^2} = \frac{b^2 + a^2}{a^2 b^2}$$



$$\frac{1}{p^2} = \frac{b^2}{a^2 b^2} + \frac{a^2}{a^2 b^2}$$

$$\frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$$

Hence proved.

QUESTION

In an Isosceles triangle OAB, O is the origin and $OA = OB = 6$. The equation of the side AB is $x - y + 1 = 0$. The area of triangle is :

(Homework Problem)

1 $2\sqrt{21}$

2 $\sqrt{142}$

3 $\frac{\sqrt{142}}{2}$

4 $\frac{\sqrt{71}}{2}$

QUESTION

A line with positive rational slope, passes through the point **A(6, 0)** and is at a distance of 5 units from **B(1, 3)**. The slope of the line is :

1 $\frac{15}{8}$

2 $\frac{8}{15}$

3 $\frac{5}{8}$

4 $\frac{8}{5}$

(Homework Problem)

$| -k | = k$ ✗
 $| k | = | k |$ ✓

Thank You